



SYNOPSIS

Privacy-Aware Audiovisual Data



PART 1 OF 5

Infrastructure for Privacy-Preserving Research

SYNOPSIS, the masking day, and the data we will work with

Babajide Owoyele

Radboud University · SYNOPSIS

MDIG Summer School · Tilburg · Thursday, 16 July 2026



PART 1 OF 5 · 10:00–10:30

Welcome to Masking Day



SYNAPSIS · MDIG 2026

Today, in one sentence

By the end of today you will **mask** real research video, **measure** how much analytic utility survives, and **decide** a masking strategy for your own data.

Mask

De-identify video with open tools.

Measure

Quantify the effect on downstream analysis.

Decide

Choose operations that fit your research.

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The day at a glance

Five parts, one pipeline



10:00–10:30

1 • SYNOPSIS & the dataset — why AV data gets stuck, what we build



10:30–10:55

2 • Why mask? — the privacy–utility dilemma



10:55–11:20

3 • The taxonomy — a vocabulary of masking operations



11:20–12:30

4 • The toolchain — Mediapipe demo, MaskAnyone install & MaskBench



13:00–14:00

5 • Your video, audio & archiving — then

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The problem: data in the drawer

Where good data goes to be forgotten

- Audiovisual data is the richest record we have of human behaviour — face, gesture, gaze, voice, interaction.
- It is also **personal data**. Faces and voices identify people directly.
- So the dataset that is **too sensitive to share** and **too valuable to delete**...does neither.
- It sits on a hard disk, a locked cabinet, a corrupted backup.



Masking is how we get it out of the drawer — responsibly.

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What SYNAPSIS is

Infrastructure for the whole life-cycle of masked research video



MaskAnyone

De-identify video:
detect, segment, mask,
export.



MaskBench

Compare pose models to
maximise analytic utility
on masked video.



Archiving

FAIR¹ sharing and
long-term deposit
(DANS).



Two runtimes

Local Docker install, or
hosted on Ponyland
(Radboud HPC).

Two ways to run today

Install MaskAnyone locally (GitHub + Docker) and set up **Ponyland** (Radboud HPC) access in parallel — plus a central **Mediapiping** demo.

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Where SYNAPSIS runs

Pilots across Dutch universities

Pilot	Data	Setting
Groningen (RUG)	Classroom & interview video, face + voice masking	Institutional VRE (in-lab)
Radboud (RTA)	Teacher-trainee classrooms; mask pupils, keep teacher	Real classrooms (in the wild)
Tilburg (SMASH-FIRE)	Emergency-service drone / incident footage	Safety, high-stakes wild
TU Delft	Multimodal / gesture (HoMI Lab)	Planning

Same toolchain, very different footage — we return to these in the case-studies companion

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The week in one arc

Masking day sits in the middle of the pipeline

Mon	Motion tracking — the tracking masking must preserve
Tue	Acoustic analysis — where audio masking connects to prosody
Wed	Pose estimation — what MaskBench evaluates on masked video
Thu	Masking & anonymisation — today
Fri	Group projects — fold masking into your own work

Everything you learned this week becomes shareable once you can mask it.

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A quick ethics primer

Before we touch a single frame



Lawful basis

Personal data needs a reason to process — consent, or public interest.²



DPIA

A data-protection impact assessment names the risk before you record.



Data minimisation

Keep only what the science needs. Masking **is** minimisation for video.²

Our pilots share a DPIA template — ask if you want it for your own study.

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What you will leave with

By 14:00 today

A masked clip

Your own footage, de-identified with MaskAnyone.

A number

A MaskBench score — which pose model keeps the **most** analytic utility on your masked clip.

A strategy

An operation you can defend for your own research.

A path to share

A route from masked clip to FAIR¹, deposited dataset.

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The running dataset

Triadic concept generation — CDR Stanford, Edelman (2011) ³

- **14 teams of 3 students** collaboratively redesigning a product.
- Recorded from **multiple camera angles**, richly gestural, multi-person interaction.
- The same corpus threads through the whole week — you already met it on Monday.

We use it because it is **hard**: exactly the masking challenge real behavioural video presents.



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Why this data is hard to mask

Many people, one frame

Detection and tracking must keep three people apart — and not swap them.

Occlusion & motion

Hands cross faces; people lean, turn, leave and re-enter the shot.

Gesture is the signal

You cannot simply black out the body — the gesture **is** the research.

Voice identifies too

Faces are not the only identifier. Audio needs a masking decision of its own.

Hold on to these four tensions — every masking choice today trades against them.

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References

Sources cited in this part

- 1 Wilkinson et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data. doi:10.1038/sdata.2016.18
- 2 European Union (2016). General Data Protection Regulation (GDPR), Regulation (EU) 2016/679. Official Journal of the EU L119.
- 3 Edelman, J. (2011). Understanding radical breaks. PhD dissertation, Stanford University.



SYNOPSIS
Privacy-Aware Audiovisual Data



PART 2 OF 5

The Privacy–Utility Dilemma

Why we mask audiovisual research data — and what is at stake

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PART 2 OF 5 · 10:30–10:55

Why Mask?



SYNAPSIS · MDIG 2026

The stranger in the background

Every old recording has one

- The bystander who walked through the frame — **never asked, never told**, captured anyway.
- Consent was given once, on paper, for a world that no longer exists.
- Datasets now live in seven places, seven jurisdictions. Every copy is a new privacy decision.



Masking exists because some things should have been done — and were not.

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The two-way mirror

Visible to research, invisible to the subject

- AV research rests on an **asymmetry**: we see the participant in full; they never see how their image travels.
- That asymmetry is what makes the data powerful — and what makes it uncomfortable.
- **Trust** is the currency masking spends: subject trusts researcher, researcher trusts the toolchain, steward trusts the policy. Every link can fail.



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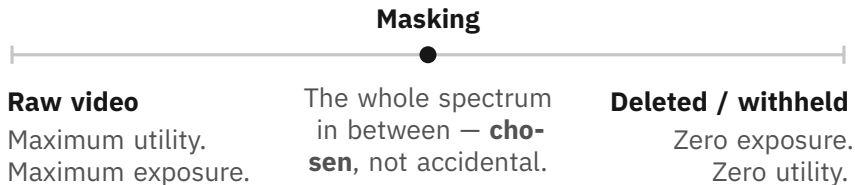
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The dilemma

Remove the identity, keep the behaviour



The question is never **whether** to trade privacy for utility — it is **where on this line** your research should sit.

What stays?

If you mask a face, what remains for science?

- The **expression** — or just its dynamics?
- The **gesture** and the reach?
- The **gaze** direction?
- The **timing** and the silence?

A good mask is a **scalpel, not a redaction**: it removes only what must go and preserves what the analysis needs.

The move that makes today work

Name what your analysis actually depends on — **before** you choose how to mask. Everything downstream follows from that answer.

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The many faces of identity

A face is not the only way to know who someone is

Direct

Face and voice — recognisable on their own.

Embodied

Gait, build, tattoos, a distinctive gesture.

Contextual

The room, a name badge, a whiteboard, a screen.

Relational

Who someone is **with** — identity by association.

Mask the face and you may still have named the person. Identity leaks from everywhere².

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The re-identifier

The card you do not want to draw — but must keep in the deck

- Today's mask meets **tomorrow's** models. A blur that fools 2026 tools may not fool 2032's ¹.
- De-anonymisation attacks recombine the fragments you left behind (gait + context + timing) ².
- Mask for the **future** adversary ², not just today's viewer — and prefer irreversible operations.

“Good enough to publish” and “good enough forever” are different thresholds.



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Consent, at scale

The contract you must now honour for every copy

- Consent was given once, to one researcher, for one study — rarely for **open sharing** or **AI training**³.
- Every new copy, jurisdiction and reuse tests that original promise again.
- Masking is how you keep the promise when the data outlives the project.



You cannot re-consent a 2011 cohort in 2026. You can mask what they never agreed to share.

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Who decides?

Masking is a shared responsibility, not a solo choice



The researcher

Names the signal the analysis needs.



The steward / DPO

Says what the law and the consent allow.



The toolchain

Executes and **records** the decision.

Ask your data steward “can I share this?” **before** you record — not after.

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And it is not optional

Ethics and law point the same way



GDPR

Faces and voices are personal data. Sharing raw AV data usually requires an explicit legal basis³.



Consent

What was agreed rarely covered open sharing or AI training.



FAIR

Funders want findable, reusable data⁴. Masking is how AV data **becomes** shareable.



The good news: masking need not gut your data.

88–95%

of pose quality (PCK) survives blurring — we prove it in Part 4.

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References

Part 2 — Why mask?

- 1 Newton, Sweeney & Malin (2005). Preserving privacy by de-identifying face images. **IEEE Transactions on Knowledge and Data Engineering**. doi:10.1109/TKDE.2005.32
- 2 Ribaric, Ariyaeenia & Pavesic (2016). De-identification for privacy protection in multimedia content: a survey. **Signal Processing: Image Communication**. doi:10.1016/j.image.2016.05.020
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SYNOPSIS
Privacy-Aware Audiovisual Data



PART 3 OF 5

The Masking Operations Taxonomy

A shared vocabulary for de-identifying audiovisual data

Babajide Owoyele

Radboud University · SYNOPSIS

MDIG Summer School · Tilburg · Thursday, 16 July 2026



PART 3 OF 5 · 10:55–11:20

The Taxonomy



SYNAPSIS · MDIG 2026

Masking is not one thing

It is a menu of operations, each with a different bargain

Operation	What it does	Identity / Signal
Blur / pixelate	Degrade the region (face or body)	Low removal, high signal
Contour / silhouette	Replace with a solid shape	Medium / medium
Remove / inpaint	Delete the person from the frame	High removal, low signal
Skeleton / pose	Keep only tracked keypoints	Removal high, motion kept
Replace / face-swap	Swap in a synthetic identity	Removal high, expression kept

The same video, five different masks — and five different research datasets. ¹

Masking is not one thing

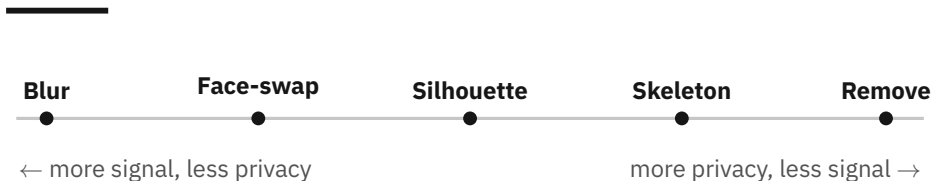
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The one axis that matters

Identity removed $\longleftrightarrow$ signal preserved



👁 No free lunch

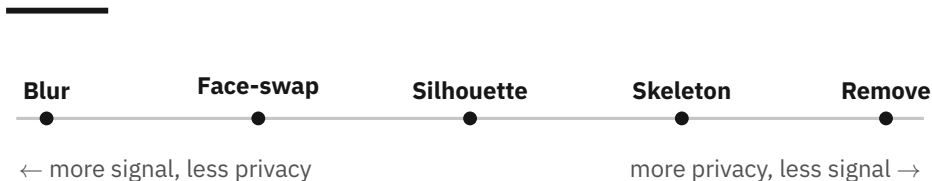
Every step right buys privacy by spending utility.

✅ The skill

Move only as far right as your privacy duty requires — no further.

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Identity removed $\longleftrightarrow$ signal preserved



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The surgeon's hand

Precision over erasure

- Blunt redaction throws away the science along with the identity.
- Precise masking removes **only** the identifying region and preserves the rest of the signal.
- Modern tools operate per-person, per-region, per-frame — not one filter over the whole video.

A scalpel that removes what must be removed, and keeps what the analysis needs.



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Choosing an operation

Start from the analysis, not the tool

Ask, in order

1. What does my analysis **measure**? (face expression? body gesture? gaze? voice?)
2. What is the minimum I can remove to meet my **privacy** duty?
3. Which operation removes the identifier **without** touching the measured signal? ⁴



Study faces?

Face-swap keeps expression, drops identity.



Study gesture?

Blur or skeleton the body; keep the kinematics.



Study talk?

Mask video, transform or redact the voice.

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Face, body, or scene?

Three regions, three different masks



The face

The strongest identifier. Blur, box, or swap it — but expression may be part of your data.³



The body

Gait, build and tattoos identify too. Silhouette or skeleton keeps the movement.



The scene

Whiteboards, screens, documents and bystanders leak identity in the background.

“Mask the face” is rarely enough — decide for **each** region.

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Can it be undone?

Reversibility and the re-identification risk

- **Reversible** operations (encryption, key-based swap) can be undone — powerful, but the key is now a liability.
- **Irreversible** operations (blur, remove) cannot be undone — safer, but you lose that data forever. ¹
- A future adversary with better models is the card you must keep in the deck: mask for **their** tools, not today's.

Rule of thumb: irreversible by default; reversible only with a governed key.



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A fourth axis: audio

Voice needs its own taxonomy

Operation	What it does	Identity / Signal
Pitch / formant shift	Alter the voice timbre	Medium / prosody kept
Voice conversion	Swap to a synthetic speaker	High / prosody partial
Redaction (bleep)	Remove segments entirely	High / content lost
Transcript-only	Release text, not audio	Highest / no acoustics

The same privacy–utility line — now along the acoustic axis (Tuesday's prosody).

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Combining operations

Real masks are mixed, per person and per region

- **Blur** the two bystanders, **skeleton** the participant whose gesture you study.
- **Swap** the face but keep the hands ²; **pitch-shift** the voice but keep the transcript.
- Modern tools apply a **different** operation to each tracked person and region — in one pass.

One video, a mask per subject: privacy where you need it, signal where you need it.



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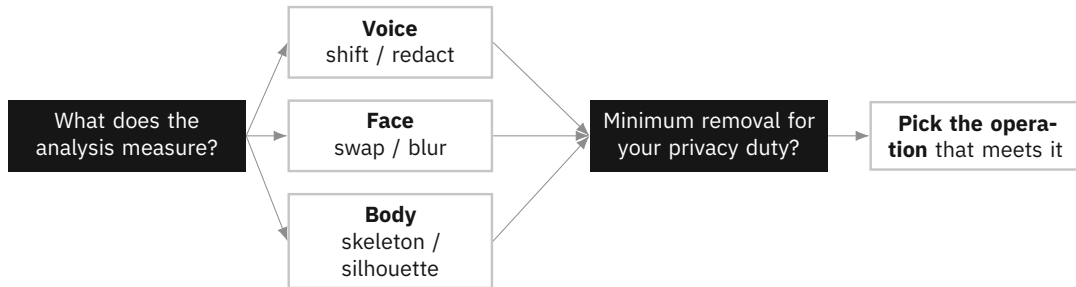
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A quick decision flow

From analysis to operation in three questions



When in doubt, mask more of the scene and let MaskBench pick the model that keeps the most signal (Part 4).

References

Sources for Part 3

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SYNOPSIS
Privacy-Aware Audiovisual Data

PART 4 OF 5

The Toolchain: MaskAnyone & MaskBench



Mask the video, then choose the pipeline that keeps the most signal

Babajide Owoyele
Radboud University · SYNOPSIS



PART 4 OF 5 · 11:20–12:05

MaskAnyone



SYNAPSIS · MDIG 2026

MaskAnyone, end to end

Open-source video masking — detect, segment, mask, export



Detection and tracking keep three people apart; you choose the operation per person.

How the hands-on runs

One central demo, then install it yourself — Ponyland access in parallel



1 • Central demo

We run **Mediapiping** together — MediaPipe full-body tracking, the pose we use in this demo.



2 • Install MaskAnyone

Walk through the GitHub + Docker install on your own laptop.



3 • Ponyland, in parallel

We set up your Radboud CLS / Ponyland access as we go — the hosted route for scale.

Two ways to run MaskAnyone: **local** today on your machine, **hosted** on Ponyland afterwards.

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Central demo: Mediapipe

MediaPipe full-body pose in Python — the pose we use in this demo

- The EnvisionBOX **Mediapipe** module:
full-body pose and landmark tracking in Python
with MediaPipe^{2 3}.
- This is the layer this demo sits on: the **skeleton**
you can mask **to**, and the keypoints MaskBench
and the gesture challenge measure.
- See the movement without the person —
masking, in its purest form.



envisionbox.org/embedded_Mediapipe.html

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Installing MaskAnyone

The MaskAnyoneProdInfrastructure repo — Docker Compose does the rest

1. **Clone** the infra repo:
`git clone github.com/MaskAnyone/
MaskAnyoneProdInfrastructure.git maskanyone`
2. **Configure:** `cp .env.dist .env` (set a password)
3. **Pull images:** `docker compose -f
docker-compose-local.yml pull`
4. **Start:** `... up -d postgres, wait 10s, then ... up -d`
5. **Open** `https://localhost` — run on a triadic clip.

GPU laptop? Swap in `docker-compose-local-gpu.yml`.

optional/advanced)

Only for the local Docker route: Docker · git · ~**50 GB** free disk · Python (for Mediapipe) · our WiFi.

No install needed for the hosted route — just a Ponyland / CLS login on the hand-out.

Your first masking job

Local or hosted, the workflow is the same

1. **Upload** a clip from the triadic dataset.
2. **Detect** — watch it separate the three team members.
3. **Choose** a masking operation (blur, silhouette, skeleton, swap).
4. **Preview** a few frames; adjust per person.
5. **Export** the masked video.

Advanced session: mixed operations, tracking fixes, and the six things that usually go wrong.

Two runtimes

Your **local** Docker instance and **Ponyland** behave the same — run small clips locally, big jobs on the HPC.



PART 4 OF 5 · 12:05–12:30

MaskBench



SYNAPSIS · MDIG 2026

Which pose model keeps the most signal?

MaskBench compares estimators to maximise analytic utility

- Run **several** pose models over the same (masked) clip and score each against the unmasked reference.⁵
- Pick the estimator that **maximises analytic utility** for your question — **before** you run any downstream analysis.
- The yardstick is **PCK** (plus RMSE and kinematics): how much of the signal each model preserves under masking.

The question is not just “did masking break it?” — it is “which pipeline keeps the most signal?”



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The headline result

Blurring keeps the body signal almost intact

✓ Pose quality retained after masking:

88–95% PCK⁵

Operation	Privacy	Pose utility (PCK)
Blur	Good	88–95%
Silhouette	Strong	Moderate
Skeleton only	Strong	Motion preserved by design
Remove	Strongest	Lost

The privacy–utility dilemma is real — but for many analyses the signal you keep is more than enough.

Provenance: masking is a claim you must be able to audit

- Record **what** was done, to **what**, by **whom**, with **which** tool, on **what** date.
- Without that trail, “masked” is a claim no reviewer, steward, or future user can check.
- MaskAnyone + MaskBench emit this record so your masked dataset is defensible.



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MaskAnyone under the hood

Four models, one job: isolate the person, then decide



YOLO

Detect every person,
frame by frame.



SAM2

Segment the exact
region per person.



Pose¹

Track keypoints for the
skeleton option.



Inpaint

Blur, replace, or remove
the region.

Why a pipeline of specialised models

Isolating each person **before** pose estimation stops the model confusing one body's limbs for another's — the key to multi-person, in-the-wild video.

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skeleton option.



Inpaint

Blur, replace, or remove
the region.

Why a pipeline of specialised models

Isolating each person **before** pose estimation stops the model confusing one body's limbs for another's — the key to multi-person, in-the-wild video.

MaskAnyone under the hood

Four models, one job: isolate the person, then decide



YOLO

Detect every person,
frame by frame.



SAM2

Segment the exact
region per person.



Pose¹

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What MaskBench actually measures

PCK is the headline — but not the only number

✓ Detection & localisation

PCK (keypoints in the right place), detection rate, **RMSE** / MPJPE against the unmasked reference.

📈 Kinematics

Velocity, acceleration, **jerk** — does the masked signal still **move** like the original?

Seven pose estimators

MaskBench runs the same masked clip through **seven** pose back-ends (incl. MaskAnyone variants) so you can **choose the best** for your data — not just trust one.⁵

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Tested where it is hard

Increasing scene complexity, up to real-world video

- MaskBench is validated on **four datasets of increasing complexity** — not just clean lab clips.
- It targets **harder** footage — from **TED talks** to multi-speaker studio scenes (TragicTalkers).
- That is the point: the benchmark meets the footage your research actually has.

Lab-grade rigour, applied to in-the-wild data — the whole SYNOPSIS bet.



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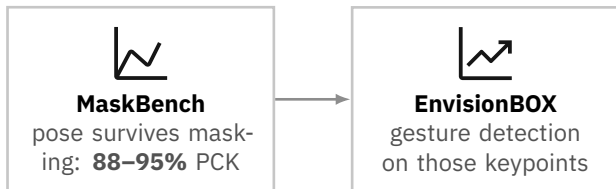
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MaskBench feeds a bigger challenge

From pose fidelity to the EnvisionBOX gesture challenge



The open gesture-detection challenge — **7 corpora, 150+ speakers, 4 languages, 32,575 clips** — runs on keypoints from **masked** video. MaskBench is what certifies they are trustworthy.

When something goes wrong

The six symptoms you will actually hit

Daemon / port

Service not up, or the port is taken — restart it.

Restart loop

A job wedged — clear it and re-run.

Out of memory

Clip too long / too big — shorten or downscale.

Detect / track drops

Person lost on occlusion — add a keyframe hint.

ID swap

Two people merged — separate the tracks.

GPU busy

Queue full on Ponyland — wait or pick a smaller model.

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Sources for this part — numbered to match the markers

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SYNOPSIS

Privacy-Aware Audiovisual Data



PART 5 OF 5

Your Video, Audio & Archiving

Bring your own footage, mask the sound, and share it FAIRly

Babajide Owoyele

Radboud University · SYNOPSIS

MDIG Summer School · Tilburg · Thursday, 16 July 2026



PART 5 OF 5 · 13:00–13:30

Bring Your Own Video



SYNAPSIS · MDIG 2026

Your turn: bring your own video

The point of the whole day

1. **Bring** a short clip — your own data, a stand-in, or fetch one with **VIDEODROP**.
2. **Mask** it with the operation you argued for in Part 3.
3. **Measure** it with MaskBench³ — did your signal survive?
4. **Discuss** the trade-off you landed on, and why.

No footage of your own? Use a second clip from the triadic dataset⁴ — or **VIDEODROP** (our local yt-dlp downloader: pick quality, split into chunks, runs on your machine).

Success looks like

A masked clip you would be willing to **share** — and a number that says how much analytic utility it **kept**.

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Audio is a face too

Multi-modal masking, briefly

- A voice identifies a speaker as surely as a face identifies a person.
- Options: **pitch/formant transformation**, full **redaction**, or **transcript-only** release.²
- Each trades privacy against prosody — the same dilemma, a different channel.
- Links straight to Tuesday's acoustic analysis: mask the identity, keep the prosodic signal.



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Dilemmask

A card-deck session for the trade-off you cannot compute

- Pull one card each from **Past, Present, Future.**
- Metaphors surface tensions; questions surface challenges; keywords name opportunities.
- Use them on **your** dataset: what did you promise, what stays, who is the next steward?



Some masking decisions are not technical.
Dilemmask is where we make them out loud.

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Bring-your-own-video checklist

So your clip actually works on the day

The clip

30–90 s, MP4/H.264, faces visible. Short beats long.

The people

Know who is under study and who is a bystander.

The permission

Only bring footage you are allowed to process.⁵

The plan

Decide the operation before you upload — from Part 3.

No clip you can share? A second angle from the triadic dataset is ready to go.

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Audio masking, in practice

Three dials, one trade-off

Transform

Pitch / formant shift: hides the speaker, keeps timing and prosody.

Redact

Bleep names and identifiers: strongest, but you lose that content.

Transcribe

Release text only: no acoustics at all — maximum privacy.

Pick per research question: studying **what** was said, or **how** it was said? ²

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PART 5 OF 5 · 13:30–13:40

Archiving & Sharing



SYNAPSIS · MDIG 2026

From masked clip to citable dataset

A deposit checklist, then DANS

The deposit checklist — four practical steps, not the FAIR principles (those are Friday's bonus deck).



Findable

Catalogued and discoverable.



Masked

Shareable because it is de-identified.



Documented

Provenance travels with the data.



Deposited

Long-term home at DANS.¹

The library that cannot be visited

One frontier is still ahead: the data need never leave home at all. More on that in a moment.

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The DANS deposit, step by step

Where a masked dataset goes to live

- 1 **Package** the masked video with its provenance manifest (what was masked, how).
- 2 **Describe** it: metadata, keywords, the consent and lawful basis.
- 3 **Set access:** open, restricted, or request-based — masking widens what you can offer.
- 4 **Deposit** at DANS; receive a DOI. Your dataset is now findable and citable.

The masked clip you made today is one manifest away from a citable record.¹

Licensing & access

Sharing is a spectrum, not a switch



Open

Masked enough to publish under a CC licence — maximum reuse.



Restricted

Access under a data-use agreement — for sensitive but shareable data.



Request-based

Vetted requests only — the data stays put, you approve each use.

The better you mask, the further left you can move — and the more your data gets used.

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The frontier: don't move the data

Federation, and where SYNAPSIS is heading

- Even masked, moving data multiplies risk.
The next step: **don't move it**.
- **Federation** — the model travels to the data; only results come back.
- Masked corpora + federation is what makes community-driven gesture-detection challenges like **EnvisionBOX** (envisionbox.org) possible without a single raw video leaving home.

Data **stays where it is**; computation travels to it.
Privacy by structure, not by promise — the library that
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Wrap-up

What to carry into your Friday group project



Masking is a choice

A point on the privacy–utility line — pick it deliberately.



Start from the analysis

Name your signal first; the operation follows.



Measure, don't assume

MaskBench turns “masked” into a number (88–95% PCK).³



Share it FAIRly

Masked + documented = a dataset you can deposit.¹

On Friday: fold a masking step into your project — and make your data shareable by design.

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Further reading

- 1 Wilkinson et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. **Scientific Data**. doi:10.1038/sdata.2016.18
- 2 Owoyele et al. (2026). MaskingOPS: A Tutorial for Masking Operations in Behavioral Research. **BRM** (under review).
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- 5 European Union (2016). General Data Protection Regulation (GDPR), Regulation (EU) 2016/679.



Thank you.

Mask it. Measure it. Share it.



SYNOPSIS

Privacy-Aware Audiovisual Data



COMPANION DECK · CASE STUDIES

Masking in the Wild

Case studies from the design lab to the incident ground

Babajide Owoyele

Radboud University · SYNOPSIS

MDIG Summer School · Tilburg · Thursday, 16 July 2026



COMPANION DECK

Case Studies



SYNAPSIS · MDIG 2026

The lab and the wild

Two kinds of data — and the bridge between them

In the lab

High control, low ecology.
Scripted, consented up front
— but ecologically thin.

In the wild

Low control, high ecology.
Real behaviour —
privacy-laden, hard to
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SYNAPSIS

The bridge: naturalistic
recordings at lab-grade
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The interesting science is in the wild. Masking is how we study it **ethically**.

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Seven settings, one toolchain

The same MaskAnyone + MaskBench pipeline, very different footage

Setting	Where	Masking challenge
Design triad (Edelman)	Lab	3 people, multi-camera; gesture is the signal
Dyadic interaction	Lab	Two speakers; alignment & prosody must survive
Teacher training (RTA)	Wild	Selective: mask pupils, keep the teacher
Emergency (SMASHFIRE)	Wild	Drone / incident footage; diverse bodies
Doctor-influencer *	Wild	Single speaker; face AUs + speech
Religious discourse *	Wild	Public figures, crowds and hinterland

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Masking Day – Case Studies

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Masking Day – Case Studies

Seven settings, one toolchain

The same MaskAnyone + MaskBench pipeline, very different footage

Setting	Where	Masking challenge
Design triad (Edelman)	Lab	3 people, multi-camera; gesture is the signal
Dyadic interaction	Lab	Two speakers; alignment & prosody must survive
Teacher training (RTA)	Wild	Selective: mask pupils, keep the teacher
Emergency (SMASHFIRE)	Wild	Drone / incident footage; diverse bodies
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CASE STUDY · LAB

The Design Triad



SYNAPSIS · MDIG 2026

The design triad

Triadic concept generation — Edelman (2011)

- **14 teams of 3**, redesigning a product, multiple camera angles.¹
- Gesture and joint attention **are** the research — you cannot black out the bodies.
- **Mask:** blur the faces, keep the skeletons; separate the three people so tracks never swap.

The controlled case — where we learn what masking costs before taking it to the wild.



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CASE STUDY · WILD

Teacher Training (RTA)



SYNAPSIS · MDIG 2026

Teacher training in real classrooms

Radboud Teachers Academy — selective masking

- Teacher-trainees recorded in live classrooms (45–90 min), processed on Ponyland HPC.
- **Selective mask:** de-identify the **pupils**, keep the **teacher** identifiable (with consent).
- Different subjects, different operations — in one pass. Consent-aware, per person.³

Masking is not all-or-nothing: it protects the many while preserving the one under study.



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CASE STUDY · WILD · HIGH STAKES

Emergency (SMASHFIRE)

Services



SYNAPSIS · MDIG 2026

Fire & rescue: SMASHFIRE

Tilburg × Brandweer Rotterdam-Rijnmond — masking in a safety context

- Drone, bodycam and incident footage — on the order of tens of thousands of incidents a year, with short retention windows.
- **Bulk mask** stored footage for sharing with partners and insurers.
- **Duty-of-care mask**: pose / silhouette models confirm a person is present without forcing the remote operator to **see** distressing imagery.
- Real tension: GDPR erasure vs **Archiefwet** retention.



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CASE STUDY · WILD

Public Figures



SYNAPSIS · MDIG 2026

In the wild: public video

Doctor-influencers, politicians, press briefings

Doctor-influencer

Single speaker, public YouTube. Facial action units + speech — mask identity, keep the expression channel.

Politicians & pressers

Crowds and bystanders in frame; consent you cannot collect after the fact. Mask the many, study the one.

Why “public” is not “free”

Being filmed in public does not make a bystander's face fair game for a shared research corpus. Masking is what makes public footage **reusable**.²

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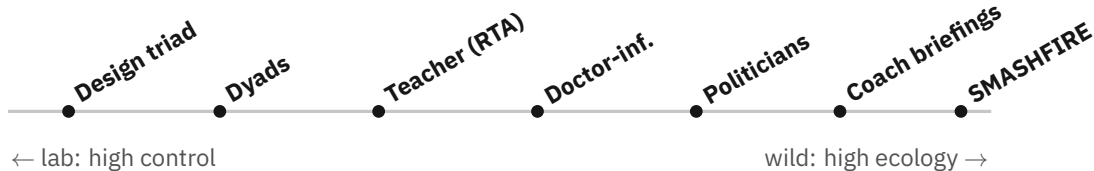
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The whole set, on one axis

Control $\longleftrightarrow$ ecology



One toolchain spans the axis — what changes is **how much** of the scene you must mask.



THE CONNECTION

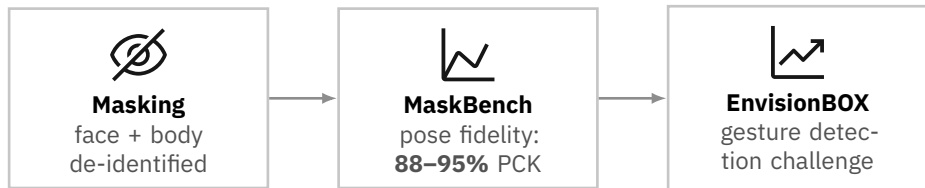
MaskBench → EnvisionBOX



SYNAPSIS · MDIG 2026

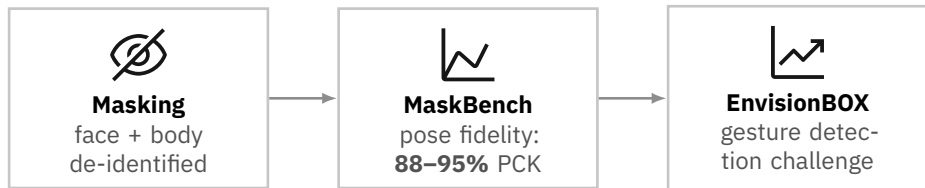
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MaskBench certifies the signal the gesture challenge consumes



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✓ The claim

MaskBench picks the pose model that preserves the most kinematic signal from **masked** video. ⁴

👤 The challenge

EnvisionBOX benchmarks gesture detectors on those keypoints: **7 corpora, 150+ speakers, 4 languages, 32,575 clips.**

Why this matters

Privacy-preserving keypoints, not privacy-destroying ones

- The gesture challenge does not need the faces — it needs the **movement**.
- MaskBench lets the community share and benchmark on **masked** corpora with confidence.
- Masking → MaskBench → EnvisionBOX is one pipeline: de-identify, certify, then compete on the science.

The open challenge is only open because the data underneath it is masked.



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Takeaways

What the case studies have in common



The wild is the point

Every high-value setting is multi-person and privacy-laden.



Selective, per subject

Mask the many, preserve the one under study.



Certify, don't assume

MaskBench turns “still usable” into a number.



Open by masking

Masked corpora are what make shared challenges possible.

One toolchain, from the design lab to the incident ground.

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References

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SYNOPSIS
Privacy-Aware Audiovisual Data



CLOSING SESSION · 30 MIN

Dilemmask

A card-deck session for the trade-offs you cannot compute

Babajide Owoyele

Radboud University · SYNOPSIS

MDIG Summer School · Tilburg · Thursday, 16 July 2026

What Dilemmask is

Some masking decisions are not technical — we make them out loud

- A **30-card**, tarot-style working deck for the Masking Lab.
- Three suits — **Past, Present, Future** — mixing **metaphors, questions, and keywords**.
- It turns the privacy–utility dilemma from a setting you compute into a conversation you have.



The number tells you what masking **keeps**. The cards tell you what it **means**.

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Session at a glance

Thirty minutes, four moves



Clock	Move	What happens
0–4 min	Frame it	What the deck is; why cards, not slides
4–7 min	Rules	Draw, read, map to the grid
7–22 min	Draw & play	One Past, one Present, one Future — place & discuss
22–27 min	On your own data	Turn the grid on the dataset you brought

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Anatomy of a card

Front names it; back reads it

Metaphor

An image-driven framing —
The Two-Way Mirror, The Surgeon's Hand. Holds two truths at once.

Question

An open prompt — “**If you mask a face, what stays?**”
No right answer, only a conversation.

Keyword

A single concept —
Provenance, Federation, Trust. Names something to build on.

Front = name, pictogram, suit. **Back** = the reading. Art is placeholder — we are testing whether it **lands**, not the final look.

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The three suits

Where each card sits in time



Past

The inherited situation — what we built on and cannot recover.

The Drawer · The Stranger in the Background · Permission · Loss



Present

The live tensions — the tools and choices in the room today.

The Mask · The Two-Way Mirror · The Surgeon's Hand · What Stays · Trust



Future

What we are building toward — the questions that outlive the project.

The Library that Cannot Be Visited · The Re-Identifier · Federation · Consent.Future

A few of the 30 — you draw one card from each suit.

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How to play

Three cards, one grid



1 • Draw

Each person pulls **one Past, one Present, one Future** card.



2 • Read

Metaphors surface **tensions**; questions surface **challenges**; keywords name **opportunities**.



3 • Map

Lay the three cards on a shared grid and talk through it as a group.

No wrong pull — the card you did not want is often the one worth discussing.

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The grid

Where the cards land — read each spread across four quadrants

Paradox / tension Metaphors — the framings that hold two truths at once.	Challenge Questions — the open problems to work on.
Opportunity Keywords — the concepts to build on.	Wildcard Anything that jumps suits or reframes the table.

Draw it big on a whiteboard or a shared sheet — the placement **is** the discussion.

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Discuss the spread

What does the Past → Present → Future line say?

- Read the three cards as **one line through time** — not three separate draws.
- Where is the **friction**? A Past permission that the Present tool cannot honour; a Future you have not consented for.
- Let a beat of silence sit after each card. The pause does the work.

The deck opens questions — it does not grade them.
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Quick variants

If you have less time, or a colder room



Single provocation

Draw **one** card, five-minute discussion. A good cold open.



Pair trade

Two people swap one card each and argue the **other's** reading.



The card you fear

Find **The Re-Identifier**. Discuss what would undo your masking.

Or run it prompt-only: ignore the metaphors, read the **questions**, three minutes each, move on.

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Two people swap one card each and argue the **other's** reading.



The card you fear

Find **The Re-Identifier**. Discuss what would undo your masking.

Or run it prompt-only: ignore the metaphors, read the **questions**, three minutes each, move on.

Facilitator tips

Holding the room

Let silence sit

Read the card aloud, then wait. The pause is the method, not dead air.

No wrong answers

The deck opens questions; it does not grade them. Follow the energy in the room.

Surface risk sideways

Use the darker cards to name failure modes without lecturing.

Confusion is a finding

If a metaphor confuses someone, that is data — note it for the feedback pass.

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On your own data

Turn the cards on the dataset you brought

- **Past:** what did you promise the person on camera — and does it still hold?
- **Present:** if you mask the face, **what stays** that your science needs?
- **Future:** who is the next steward, and can they audit what you did?

Carry the grid into your Friday project — the decisions you make here are the ones no tool makes for you.



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This is a prototype

We are testing whether the cards land — not the final look

- Art is placeholder pictograms; some copy is still draft.
- At the summer school we are testing **whether the cards land** — which spark discussion, which fall flat.
- Your reactions in this room feed straight back into the deck.

Feedback goes back into the card sources, then we regenerate the print deck.



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Capture feedback

What to watch for, and jot down



Watch for

The question behind it

Cards that land

Which sparked the best discussion? Which fell flat or got skipped?

Confusing metaphors

Any card where the metaphor misfired or the term stayed opaque.

Missing / wrong

A concept the deck should cover but doesn't; anything factually off (GDPR, FAIR, anonymisation).

Timing & flow

Did the play run to time? Too many cards? Too few?

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Three questions for testers

Before you leave the table

1 • Keep

Which card would you **keep**?

2 • Cut

Which would you **cut** or
rewrite?

3 • Use

Would you use this deck in
your own teaching or lab —
and **when**?

Why we ask

The cards only earn a place in the deck if a facilitator would reach for them again.
Your three answers are the whole test.

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Carry the grid into Friday

The decisions no tool makes for you

- Dilemmask turns the privacy–utility dilemma from a setting you **compute** into a conversation you **have**.
- Take your Past → Present → Future line into your own project.
- The tool gives you a number. The table gives you the meaning.

Some masking decisions are not technical — we make them out loud.



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